



Potentiometer

Hardware Overview



What is it?

The **Potentiometer** is a rotating knob, like a volume dial. It is an **Analog Input** device. On K-Comp, you can use it to set a precise value, like controlling the speed of a game, or the frequency of a sound.



Electrical Explanation

Inside the metal housing is a resistive strip. The knob controls a **Wiper** that slides along this strip.

- Turning the knob changes the length of the strip the electricity must flow through.
- This creates a variable **Voltage Divider**.
- The output voltage changes from **0V** to **5V** depending on the knob's angle.

Key Hardware Facts:

- **Connection:** The wiper is connected to **Analog Pin A0** on the microcontroller.
- **Function:** It converts physical rotation (angle) into electrical voltage.
- **Signal Flow:** Rotation → Resistance Change → Voltage Change (0-5V) → ADC → Number.



How Coordinates Work: Analog to Digital (ADC)

The processor cannot understand "Voltage". It only understands numbers. The **ADC (Analog-to-Digital Converter)** takes the 0-5V signal and chops it into

Knob Position	Voltage	Digital Value
Far Left	0V	0
Middle	2.5V	~512
Far Right	5V	1023

1024 steps. Standard analog inputs read a value from 0 to 1023. However, the **K-Comp library** automatically scales it to a range range from 0 to 100.

Software & Programming



kcomp_potentiometer.h

To use the knob, you must include the library at the top of your sketch: `#include <kcomp_potentiometer.h>`.

Function Reference

Function Name	Description & Parameters
<code>void initPotentiometer()</code>	Setup: Initializes the hardware connection on Pin A0. This must be called once inside your <code>setup()</code> function.
<code>int getPotentiometerRaw()</code>	Read Raw Value: Returns the precise position of the knob. Returns: An integer between 0 (Min) and 1023 (Max).
<code>int getPotentiometerValue()</code>	Read Percentage: Returns the position as a simple percentage. Returns: An integer between 0 and 100. <i>(Useful for things like screen brightness or volume).</i>

⚠ Pitfall: "Signal Jitter" (Noise)

Analog sensors are never perfectly stable. Even if you don't touch the knob, the value might wiggle slightly (e.g., jumping between 500, 501, and 499).

- **The Solution:** If you need a perfectly steady number, you might need to ignore small changes in your code (e.g., "only update if the value changes by more than 2").

**Sample Program**

This program prints the direction and intensity to the Serial Monitor.

```
#include <kcomp_potentiometer.h> // 1. Include the library

void setup() {
    Serial.begin(115200);
    initPotentiometer(); // 2. Initialize the hardware
    Serial.println("Potentiometer Ready!");
}

void loop() {
    // 3. Read the raw precision value (0-1023)
    int raw = getPotentiometerRaw(); //

    // 4. Read the simple percentage value (0-100)
    int percent = getPotentiometerValue(); //

    Serial.print("Raw: ");
    Serial.print(raw);
    Serial.print(" | Percent: ");
    Serial.println(percent);

    delay(500); // Wait 0.5s so the screen doesn't scroll too fast
}
```

💡 **Try it out:** You can find the full sample program **SimplePotentiometer** in the K-Comp examples folder!